Equivariant Systems Theory and Observer Design for Autonomous Systems

R. Mahony J. Trumpf T. Hamel

Australian National University
UNIVERSITÉ CÔTE D'AZUR

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Outline

1. Homography example
   - What is a Homography
   - Homography Kinematics
   - Observer design

2. Practical Session
   - Homography versus Rigid Body velocity
   - Observer Design
   - Experiments

Outline

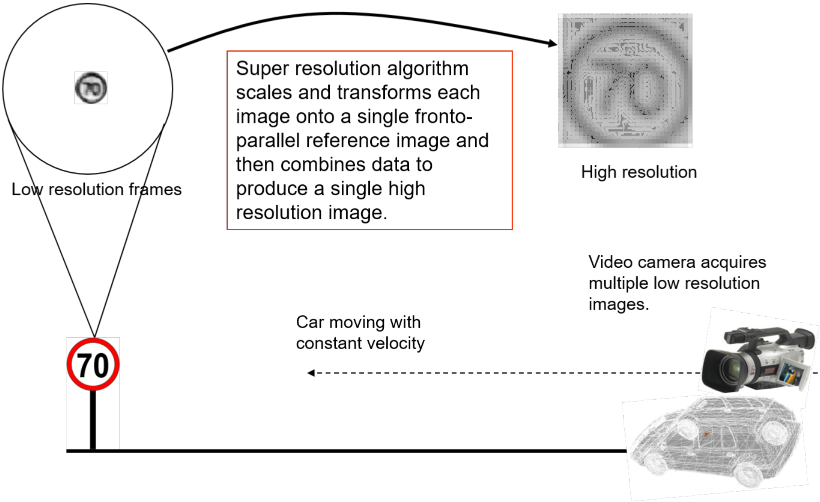
What is a homography?

A homography is an invertible mapping relating two images of the same planar scene.
Any two images of the same planar surface in space are related by a homography.

It has many practical applications, such as:

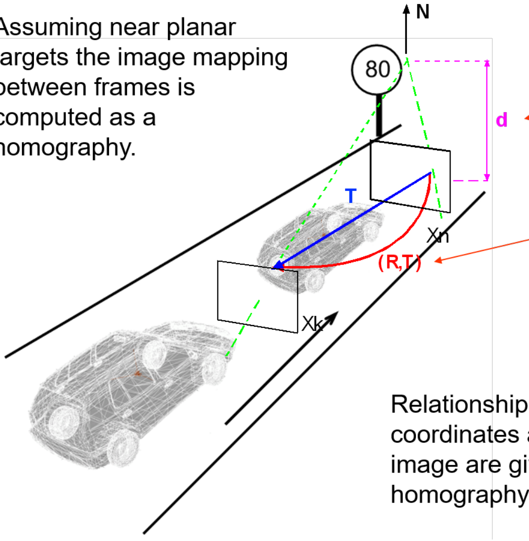
- Image rectification,
- Image registration,
- Visual metrology,
- Stereo vision,
- Scene understanding (Image mosaicing),
- Computation of camera motion between two images,
- ...

Super Resolution Example 1



Super Resolution Example 2

Assuming near planar targets the image mapping between frames is computed as a homography.



Normal distance between street sign planar surface and reference frame.

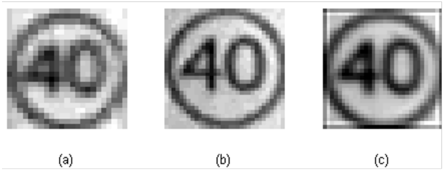
Rigid-body transformations between image reference frames.

Relationship between image coordinates and the reference image are given by a separate homography for each of the images.

Super Resolution Example 3

Typical image extracted out of video Sequence.

An FPGA system searches for speed signs using a circle matching algorithm, computes a bounding box and extracts low resolution (LR) images of the speed sign in real time.

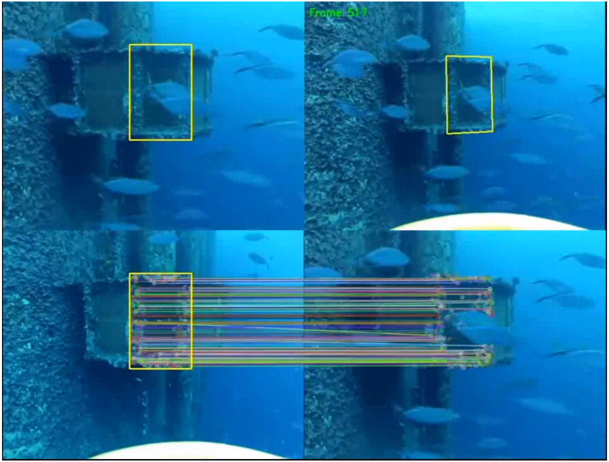


LR frame | Bi-Linear | SR Image

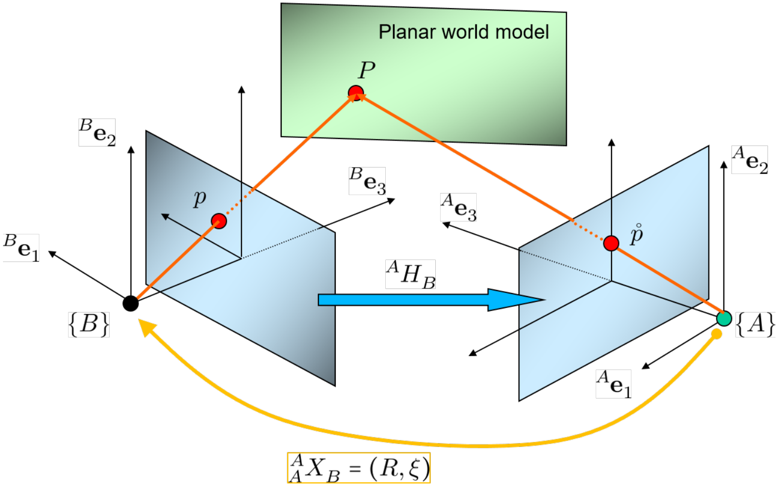
Once the data is warped onto a single reference image then all the LR data can be combined together to form a high resolution image.

The simplest method is bi-linear interpolation.
Super-Resolution methods tend to use statistical criteria to obtain better results.

Homography Tracking



The geometry of homographies



What is a homography?

Assume a calibrated camera and that there is a planar surface π containing a set of n target points ($n \geq 4$) so that

$$\pi = \left\{ {}^{B}P \in \mathbb{R}^3 : {}^{B}\eta^\top {}^{B}P - {}^{B}d = 0 \right\},$$

where ${}^{B}\eta$ is the unit normal to the plane expressed in $\{B\}$ and ${}^{B}d$ is the distance of the plane to the origin of $\{B\}$.
Consider the Coordinate mapping between frames of reference ${}^{A}_{A}X_B = (R, \xi)$. Then

$$R\,{}^{B}P_i + \xi = {}^{A}P_i$$

$$R\,{}^{B}P_i + \frac{\xi\,{}^{B}\eta^\top\,{}^{B}P_i}{{}^{B}d} = {}^{A}P_i$$

$$\left(R + \frac{\xi\,{}^{B}\eta^\top}{{}^{B}d} \right) {}^{B}P_i = {}^{A}P_i$$

Homogeneous coordinates

Define a matrix H by

$$H := \left(R + \frac{\xi\, {}^B\eta^\top}{{}^B d} \right)$$

Then

$$\left(\frac{{}^B\mathbf{e}_3{}^\top\, {}^BP_i}{{}^B\mathbf{e}_3{}^\top\, {}^BP_i} \right) H\, {}^BP_i = \left(\frac{{}^A\mathbf{e}_3{}^\top\, {}^AP_i}{{}^A\mathbf{e}_3{}^\top\, {}^AP_i} \right) {}^AP_i$$
$$\left({}^B\mathbf{e}_3{}^\top\, {}^BP_i \right) Hp_i = \left({}^A\mathbf{e}_3{}^\top\, {}^AP_i \right) \mathring{p}_i$$
$$Hp_i = \left(\frac{{}^A\mathbf{e}_3{}^\top\, {}^AP_i}{{}^B\mathbf{e}_3{}^\top\, {}^BP_i} \right) \mathring{p}_i$$

$$Hp_i \cong \mathring{p}_i$$

Here we use $\cong$ to denote equality up to scale. This equivalence can also be associated with homogeneous coordinates in 2-dimensions.

Homography operator

The matrix H

$$H := \left(R + \frac{\xi^B \eta^\top}{^B d} \right)$$

defines a *homography* mapping from one image B to A

$$Hp \cong H \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} \cong \begin{pmatrix} \mathring{u} \\ \mathring{v} \\ 1 \end{pmatrix} \cong \mathring{p}.$$

Since scale is not considered then any

$$H' = \sigma H$$

for $\sigma \neq 0$ will generate the same homography mapping.
The set of possible homographies corresponds to an 8-dimensional matrix sub-manifold of $\mathbb{R}^{3\times 3}$.

Group coordinates for homographies

Homography versus element of the Special Linear Goup $SL(3)$

$$SL(3) = \{S \in \mathbb{R}^{3\times 3} \mid \det S = 1\}.$$

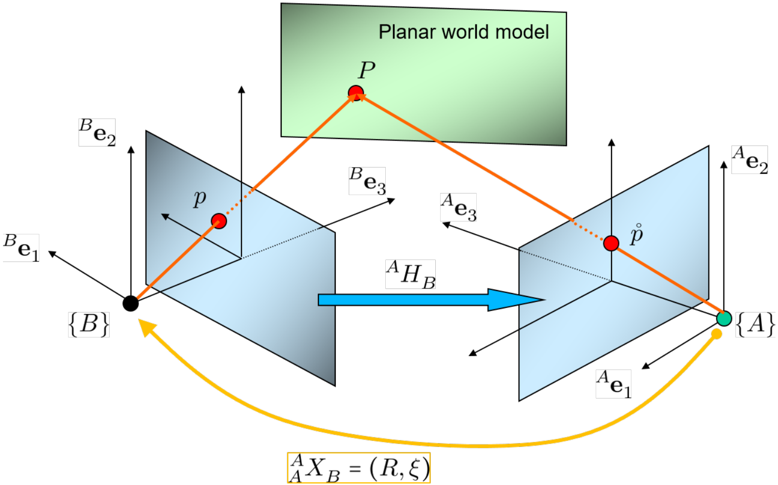
Any homography matrix is associated with a unique matrix $\bar{H} \in SL(3)$ by re-scaling

$$\bar{H} = \frac{1}{\det(H)^{\frac{1}{3}}} H$$

such that $\det(\bar{H}) = 1$.

The Lie group $\mathrm{SL}(3)$ is another classical Lie group like $\mathrm{SE}(3)$ and $\mathrm{SO}(3)$. We use it here to provide *group coordinates* for the space of image homographies.

The geometry of homographies



Nonlinearity of the Homography mapping

Although the homography mapping

$$p \mapsto \mathring{p} \cong Hp$$

appears to be linear, the rescaling invariance operation in fact introduces nonlinearity.
Consider spherical image projections, that is $p_i = \frac{1}{|{}^B P_i|} {}^B P_i \in \mathrm{S}^2$ and $\mathring{p}_i = \frac{1}{|{}^A P_i|} {}^A P_i \in \mathrm{S}^2$.
Then the map

$$\rho \colon SL(3) \times \mathrm{S}^2 \longrightarrow \mathrm{S}^2,$$
$$(H, \mathring{p}) \mapsto \rho(H, \mathring{p}) := \frac{H^{-1}\mathring{p}}{|H^{-1}\mathring{p}|}$$

describes the homography operation.

Group action

Consider the operator

$$\rho\colon SL(3) \times \mathrm{S}^2 \longrightarrow \mathrm{S}^2,$$
$$(H, \mathring{p}) \mapsto \rho(H, \mathring{p}) := \frac{H^{-1}\mathring{p}}{|H^{-1}\mathring{p}|}$$

One has

$$\rho(H_1, \rho(H_2, \mathring{p})) = \rho(H_2 H_1, \mathring{p}),$$

and also the identity property

$$\rho(I, \mathring{p}) = \mathring{p}$$

These equations are characteristic of a structure known as 'right' group action. In this case ρ is a right group action of $SL(3)$ on S^2.

Measurement

In particular,

$$p_i = \frac{H^{-1}\mathring{p}_i}{|H^{-1}\mathring{p}_i|}$$

This output equation is analogous to the attitude output equation

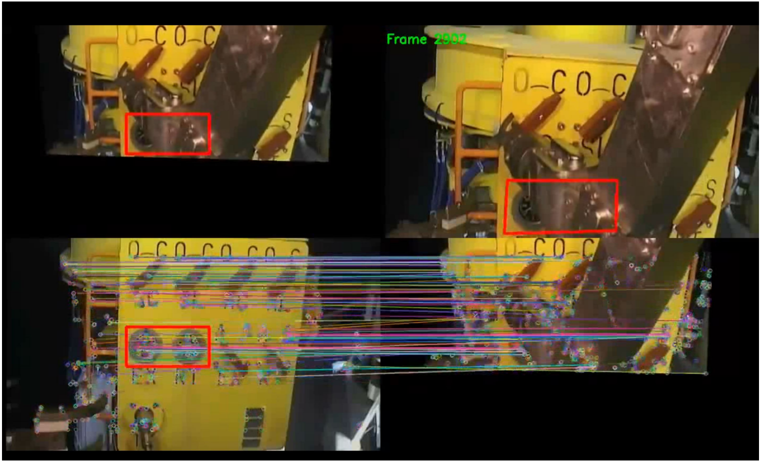
$$y_i = R^\top \mathring{y}_i$$

Thus, if we do point matching in two images of the same planar scene we obtain point correspondences $(p_i, \mathring{p}_i)$ in images B and A that can act as measurements for an observer design.

The object that we can observe is the Homography H.

$$H := \frac{1}{\det(\sim)^{1/3}}\left(R + \frac{\xi^B \eta^\top}{{}^B d}\right)$$

Homography estimation



Homography Kinematics

We postulate kinematics on the group

$$\dot{H} = HU$$

for a suitable *velocity* matrix U. We require U be chosen such that $\frac{\mathrm{d}}{\mathrm{d}t}\det(H) = 0$. That is

$$\frac{\mathrm{d}}{\mathrm{d}t}\det(H) = \det(H)\,\mathrm{tr}(H^{-1}\dot{H}) = 0$$

Thus, setting $\dot{H} = HU$ this is equivalent to

$$\mathrm{tr}(U) = 0$$

Define,

$$\mathfrak{sl}(3) = \{U \in \mathbb{R}^{3\times 3} \mid \mathrm{tr}(U) = 0\}$$

Note that $\dim(\mathfrak{sl}(3)) = 8$. $\mathfrak{sl}(3)$ is the Lie algebra of $\mathrm{SL}(3)$.

Homography kinematics

We saw that solutions to $\dot{H} = HU$, $U \in \mathfrak{sl}(3)$, preserve the group structure of $\mathrm{SL}(3)$

Show that solutions of $\dot{H} = UH$ for $U \in \mathfrak{sl}(3)$ also preserve the constraint $\det(H) = 1$.

Tangent spaces and Riemannian metric

On $\mathfrak{sl}(3)$ we place an inner product

$$\langle U, V \rangle = \mathrm{tr}(U^\top V)$$

The right invariant tangent space of $\mathrm{SL}(3)$ is given by

$$T_H \mathrm{SL}(3) = \{UH \mid U \in \mathfrak{sl}(3)\}$$

This induces a natural isomorphism from $\mathfrak{sl}(3)$ to $T_H\mathrm{SL}(3)$.

On an arbitrary tangent space we place a metric

$$\langle UH, VH \rangle_H = \mathrm{tr}(U^\top V)$$

This is known as the right invariant metric on the Lie group.

Problem set up

- Define the estimate measurement

$$\hat{p}_i = \frac{\hat{H}^{-1}\mathring{p}_i}{|\hat{H}^{-1}\mathring{p}_i|}.$$

- Define the group error

$$E = \hat{H}H^{-1}$$

- Define the measurement error

$$e_i = \frac{\hat{H}H^{-1}\mathring{p}_i}{|\hat{H}H^{-1}\mathring{p}_i|} = \frac{\hat{H}p_i}{|\hat{H}p_i|}$$

- Assume that the group velocity $U \in \mathfrak{sl}(3)$ is known,
- Assume that there is a measurement p in the target image at each time that corresponds to the point $\mathring{p}$ in the reference image.

Problem set up

- ¿ What are the observer Kinematics ?
- ¿ Compute the error kinematics ?
- ¿ Write down a good cost function ℓ? Remember you must be able to measure or know all variables in the cost function.
- ¿ Compute the gradient of ℓ?
- ¿ Use the gradient to define an innovation for the observer?
- ¿(Sketch) Prove Lyapunov stability of the error kinematics

Nonlinear observer on $SL(3)$ based on direct measurements

- Choose the estimator filter:

$$\dot{\hat{H}} = \hat{H}U + k_P \Delta \hat{H}, \quad \Delta \in \mathfrak{sl}(3)$$

- Error Kinematics $\dot{E} = -k_P \Delta E$.
- Cost function

$$\ell_{\mathring{p}_i}(E, \mathring{p}_i) = \frac{1}{2} \sum_{i=1}^{n} \left| \frac{E\mathring{p}_i}{|E\mathring{p}_i|} - \mathring{p}_i \right|^2 = \frac{1}{2} \sum_{i=1}^{n} |e_i - \mathring{p}_i|^2$$

  locally a positive definite function of E.
- Innovation

$$\Delta = \sum_{i=1}^{n} k_i (I - e_i e_i^\top) \mathring{p}_i e_i^\top$$

Problem set up

- ¿ What are the observer Kinematics ?
- ¿ Compute the error kinematics ?
- ¿ Write down a good cost function ℓ? Remember you must be able to measure or know all variables in the cost function.
- ¿ Compute the gradient of ℓ?
- ¿ Use the gradient to define an innovation for the observer?
- **Excercise: (Sketch) Prove Lyapunov stability of the error kinematics**

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Homography velocity

In practice, there is no physical sensor available that measures the homography velocity.
The full homography velocity allows for arbitrary motion of the scene as well as the camera. If we assume that the scene is static, then the homography velocity is closely related to the rigid body velocity of the camera.

$$H \cong \left(R + \frac{\xi\, {}^B\eta^\top}{{}^B d} \right)$$

where

$$\dot{R} = R\Omega_\times, \quad \dot{\xi} = RV, \quad V = {}^B_A V_B, \quad \Omega = {}^B_A\Omega_B, \quad R = {}^A_A R_B$$

One has

$$U = \left(\Omega_\times + \frac{V\, {}^B\eta^\top}{{}^B d} - \frac{{}^B\eta^\top V}{3\, {}^B d} I \right)$$

Homography velocity: some useful identities

In order to prove

$$U = \left(\Omega_\times + \frac{V {}^B\eta^\top}{{}^Bd} - \frac{{}^B\eta^\top V}{3\,{}^Bd} I \right)$$

Hint 1:

$$\begin{aligned} {}^B\eta &= R^\top {}^A\eta \qquad\qquad \eta = {}^B\eta \\ {}^B\dot{\eta} &= -\Omega \times {}^B\eta \end{aligned}$$

Hint 2:

$$ {}^Bd = {}^Ad - {}^A\eta^\top \xi$$

Hint 3:
Note that for any matrix $A \in \mathbb{R}^{n\times n}$ and $\Delta \in \mathfrak{sl}(3)$ then

$$\mathrm{tr}(A\Delta) = \mathrm{tr}\left(\left(A - \frac{\mathrm{tr}(A)}{3} I \right) \Delta \right).$$

Practical solution

In practice, even measuring the linear velocity of a camera can be difficult. However, measuring the angular velocity is easy.
Approach:

- Measure angular velocity and use this in the estimator.
- Assume the linear velocity is constant *in the body-fixed frame* and treat this as a constant bias!
- Build the observer with a bias estimator.

Observer with partially known velocity of the rigid body

Split the velocity into measured and unmeasured components

$$U := (\Omega_\times + \Gamma)$$

Assume the unmeasured component $\Gamma = \left(\frac{V\eta^\top}{d} - \frac{\eta^\top V}{3d} I\right)$ is constant.

Use the measurements Ω where available.

The resulting problem is treated as a biased velocity measurement

$$\dot{\hat{H}} = \hat{H}\left(\Omega_\times + (\hat{\Gamma}_1 - \frac{1}{3}\operatorname{tr}(\hat{\Gamma}_1)I)\right) + k_P \Delta \hat{H},$$

$$\dot{\hat{\Gamma}}_1 = \hat{\Gamma}_1 \Omega_\times + k_I \operatorname{Ad}_{\hat{H}^T} \Delta$$

$$\Delta = \sum_{i=1}^{n} k_i (I - e_i e_i^\top) \mathring{p}_i e_i^\top$$

What you will do next!